Round 2 Model and Data Guide

Supplied classifier and synthetic inspection samples

What this pack provides

Model BIT-PLATE-CV-1.0 is a task-specific pretrained binary classifier for the fictional product P-PLATE01, Demonstration metal plate. It uses image features and a learned logistic regression model. It is not a neural network or a general-purpose factory inspection system.

The model was fitted before release on 1,200 procedurally generated images, with 600 images per class. The student set contains 60 independently generated images, 30 Normal and 30 Defective. No student sample image was used to fit the model or feature standardization. Samples are development data, not a blind benchmark.

Item	Value
Product ID	P-PLATE01
Class 0	Normal
Class 1	Defective
Decision rule	Defective when defect_probability >= 0.5; otherwise Normal
Confidence shown	Probability assigned to the predicted class, from 0 to 1
Weights	Model/classifier.npz; NumPy numeric arrays only
Runtime	Python with NumPy and Pillow; CPU inference, no network call
Reference helper	Model/inference.py; keep preprocessing and threshold unchanged

Image and label conventions

All 60 sample images are 256 by 256 pixels, in PNG or JPEG format. The plate is centered against a dark border. Labels are in Samples/labels.csv; relative_path is relative to Samples. Filenames are opaque identifiers. Use label records only for computing metrics, never as an inference input.

Normal images contain surface texture and lighting variation. Defective images contain one or more simulated scratches or pits. Only the binary label is supplied and assessed. The model does not identify defect subtype, location, severity, cause, or future failure.

Data provenance and limits

Blessed IT Solution Pvt Ltd prepared these procedural illustrations for the hackathon. They are not photographs, client data, or images of actual manufactured parts. They are authorized for event use and related student submissions. No external dataset license or image download is needed.

Fixed framing is part of the model contract. Real images, rotated or displaced products, different backgrounds, and unfamiliar defects are outside its validated scope. Scores are uncalibrated model outputs, not verified probabilities of product safety. Human review remains separate.

Preprocessing Runtime and Evaluation

Exact inference pipeline

The reference helper applies EXIF orientation, converts to RGB, resizes to 128 by 128 with bilinear interpolation, converts to grayscale, and takes rows and columns 16 through 111 inclusive. Values are divided by 255. The crop excludes the surrounding border.

It extracts 17 features: mean and standard deviation; eight quantiles of median-centered intensity at 1, 5, 10, 25, 75, 90, 95, and 99 percent; five fractions below median by more than 0.03, 0.06, 0.10, 0.15, and 0.20; and mean absolute adjacent-pixel differences on both axes. Training means and scales standardize features before the learned linear score and sigmoid.

Model/inference.py is the authoritative executable preprocessing specification. Import predict and pass the saved image path. Its result contains predicted_class, confidence, defect_probability, and model_version. The application must validate uploads before calling it.

Local smoke check

From the extracted pack folder, run these commands in your prepared Python environment:

python -m pip install -r Model/requirements.txt

python Model/inference.py Samples/images/inspection_001.png

The command prints a JSON prediction without downloading weights or using a GPU. The provided requirements cover inference only; your application also needs the event stack. Model access requires no API key. The environment should be prepared before Round 2.

Metric definitions

Use Defective as the positive class. TP = Defective predicted Defective; FN = Defective predicted Normal; FP = Normal predicted Defective; TN = Normal predicted Normal. Show a 2 by 2 matrix with rows as actual labels and columns as predicted labels, both ordered Normal, Defective.

Defect recall = TP / (TP + FN). Compute it from original model outputs on all 60 labeled samples, before any human review. Report missing or failed predictions explicitly rather than excluding them silently. If no false negatives occur, state that and explain why this small synthetic sample cannot establish real-world reliability.

Validation files and model integrity

Validation/not_an_image.jpg is text with a misleading extension. Validation/corrupt_image.png is a truncated PNG. Validation/too_small.png is a valid 32 by 32 image, below the minimum dimension. These are intentional test inputs and are excluded from sample metrics.

SHA256SUMS.txt lists checksums for the model, helper, metadata, labels, and sample images. Preserve the model version with every completed inspection. Retry logic, if added, must not overwrite the original completed prediction or remove supervisor review history.